

Problem_6: The following Table shows the output (Q), the labour input (L) and capital input (K) of 15 firms of a chemical industry.

Firm	Output (1000 tons)	Labour (hour)	Capital (machine hour)
	Q	L	K
1	6	110	30
2	12	120	40
3	19	143	42
4	25	150	40
5	30	152	51
6	36	162	59
7	38	180	60
8	43	182	63
9	44	180	61
10	49	175	63
11	50	195	85
12	52	196	90
13	54	183	98
14	41	190	90
15	35	150	80

(a) Fit the Cobb-Douglas (CD) production model specified as

$$Q = \beta_0 L^{\beta_1} K^{\beta_2} e^U$$

to the above data and test the goodness of fit.

(b) Find the estimates of $\widehat{\beta}_1$ and $\widehat{\beta}_2$ and their standard errors.

(c) Test the significance of $\widehat{\beta}_1$ and $\widehat{\beta}_2$.

(d) Fit the Constant Elasticity of Substitution (CES) production model specified as

$$Q = A[\alpha L^{-\rho} + (1 - \alpha)K^{-\rho}]^{-\frac{1}{\rho}}$$

to the above data and test the goodness of fit.

(e) Find the estimates of $\widehat{\beta}_1$, $\widehat{\beta}_2$ and $\widehat{\beta}_3$ and their standard errors.

(f) Test the significance of $\widehat{\beta}_1$, $\widehat{\beta}_2$ and $\widehat{\beta}_3$.

Solution: The Cobb-Douglas production model is

$$\begin{aligned}
 Q &= \beta_0 L^{\beta_1} K^{\beta_2} e^U \\
 \Rightarrow \log_e Q &= \log_e \beta_0 + \beta_1 \log_e L + \beta_2 \log_e K + U \\
 \Rightarrow Y &= \alpha_0 + \beta_1 X_1 + \beta_2 X_2 + U
 \end{aligned}$$

where $Y = \log_e Q$, $\alpha_0 = \log_e \beta_0$, $X_1 = \beta_1 \log_e L$ and $X_2 = \beta_2 \log_e K$

$$\begin{aligned}
 \Rightarrow Y_i &= \alpha_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + U_i, i = 1, 2, \dots, 15. \\
 \Rightarrow \bar{Y} &= \alpha_0 + \beta_1 \bar{X}_1 + \beta_2 \bar{X}_2 + \bar{U}
 \end{aligned}$$

From the above two equations, we get

$$y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + u_i \quad (\text{in deviation form})$$

where $y_i = Y_i - \bar{Y}$, $x_{1i} = X_{1i} - \bar{X}_1$, $x_{2i} = X_{2i} - \bar{X}_2$

In matrix notation, we have, $y = x\beta + u$

$$\text{where, } y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{15} \end{bmatrix}, x = \begin{bmatrix} x_{11} & x_{21} \\ x_{12} & x_{22} \\ \vdots & \vdots \\ x_{1,15} & x_{2,15} \end{bmatrix}, \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}, u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{15} \end{bmatrix}$$

Now applying the OLS, we get $\hat{\beta} = (x^T x)^{-1} x^T y$

$$x^T x = \begin{bmatrix} \sum_{i=1}^{15} x_{1i}^2 & \sum_{i=1}^{15} x_{1i} x_{2i} \\ \sum_{i=1}^{15} x_{1i} x_{2i} & \sum_{i=1}^{15} x_{2i}^2 \end{bmatrix}$$

$$x^T y = \begin{bmatrix} \sum_{i=1}^{15} x_{1i} y_i \\ \sum_{i=1}^{15} x_{2i} y_i \end{bmatrix}$$

Firm	$Y = \log_e Q$	$X_1 = \log_e L$	$X_2 = \log_e K$	X_1^2	X_2^2	$X_1 X_2$	$X_1 Y$	$X_2 Y$
1	1.79	4.70	3.40					
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								

Now we calculate the mean deviation about mean,

$$\sum_{i=1}^{15} x_{1i}^2 = \sum_{i=1}^{15} X_{1i}^2 - n\bar{X}_1^2 = ???$$

$$\sum_{i=1}^{15} x_{2i}^2 = \sum_{i=1}^{15} X_{2i}^2 - n\bar{X}_2^2 = ???$$

$$\sum_{i=1}^{15} x_{1i}x_{2i} = \sum_{i=1}^{15} X_{1i}X_{2i} - n\bar{X}_1\bar{X}_2 = ???$$

$$\sum_{i=1}^{15} x_{1i}y_i = \sum_{i=1}^{15} X_{1i}Y_i - n\bar{X}_1\bar{Y} = ???$$

$$\sum_{i=1}^{15} x_{2i}y_i = \sum_{i=1}^{15} X_{2i}Y_i - n\bar{X}_2\bar{Y} = ???$$

$$(x^T x)^{-1} = ???$$

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = ???$$

$$\hat{\alpha}_0 = \bar{Y} - \hat{\beta}_1\bar{X}_1 - \hat{\beta}_2\bar{X}_2$$

$$\text{The fitted line is } Q = \hat{\beta}_0 L^{\hat{\beta}_1} K^{\hat{\beta}_2} = ???$$

Now, $SS_Y = \sum_{i=1}^{15} Y_i^2 - n\bar{Y}^2 = ???$

$SS_R = \hat{\beta}^T x^T y = ???$

$SS_E = SS_Y - SS_R$

$R^2 = SS_R / SS_Y$

Comment: Since the value of $R^2 = ???$, then we can say that the fit is

ANOVA Table

Source of variation	DF	SS	MSS	F _{cal}	F _{tab}
Regression					
Residual					
Total					

Now we have to test the following hypothesis

H_0 : There is no regression effect

H_1 : H_0 is not true

Comment: Here the tabulated value of F at 5% level of significance withanddf is and the calculated value of F is, Since the $F_{cal} > F_{tab}$ or $F_{cal} < F_{tab}$, therefore we reject or accept our null hypothesis. i.e. Regression effect is significant or insignificant.

(b) We know that $V(\hat{\beta}) = \hat{\sigma}^2 (x^T x)^{-1}$ where, $\hat{\sigma}^2$ is the MSS Residual

$$V(\hat{\beta}_1) = \hat{\sigma}^2 a_{11} = ???$$

$$V(\hat{\beta}_2) = \hat{\sigma}^2 a_{22} = ???$$

and $SE(\hat{\beta}_1) = \sqrt{V(\hat{\beta}_1)} = ???$ and $SE(\hat{\beta}_2) = \sqrt{V(\hat{\beta}_2)} = ???$

(c) We want to test the following hypothesis

$H_0: \beta_1 = 0$

Vs $H_1: \beta_1 \neq 0$

The test statistic is $t = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)} \sim t_{n-k}$

Comment:

Again, to test the following hypothesis

$$H_0: \beta_2 = 0$$

$$\text{Vs } H_1: \beta_2 \neq 0$$

$$\text{The test statistic is } t = \frac{\widehat{\beta}_2 - \beta_2}{SE(\widehat{\beta}_2)} \sim t_{n-k}$$

Comment:

(d) The CES production model is

$$Q = A[\alpha L^{-\rho} + (1 - \alpha)K^{-\rho}]^{-\frac{1}{\rho}} e^U$$

After some algebraic manipulation, we get

$$\begin{aligned} \log_e Q &= \log_e A + \alpha \log_e L + (1 - \alpha) \log_e K - \frac{\rho}{2} \alpha (1 - \alpha) (\log_e L - \log_e K)^2 + U \\ \Rightarrow Y &= \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + U \end{aligned}$$

$$\begin{aligned} \text{where } Y &= \log_e Q, \beta_0 = \log_e A, \beta_1 = \alpha, X_1 = \log_e L, \beta_2 = 1 - \alpha, X_2 = \log_e K, \\ \beta_3 &= -\frac{\rho}{2} \alpha (1 - \alpha), X_3 = (\log_e L - \log_e K)^2 \end{aligned}$$

$$\Rightarrow Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + U_i, i = 1, 2, \dots, 15.$$

In deviation form, we get

$$y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + u_i$$

$$\text{where } y_i = Y_i - \bar{Y}, x_{1i} = X_{1i} - \bar{X}_1, x_{2i} = X_{2i} - \bar{X}_2, x_{3i} = X_{3i} - \bar{X}_3$$

$$\text{Now applying the OLS, we get } \widehat{\beta} = (x^T x)^{-1} x^T y$$

$$x^T x = \begin{bmatrix} \sum_{i=1}^{15} x_{1i}^2 & \sum_{i=1}^{15} x_{1i}x_{2i} & \sum_{i=1}^{15} x_{1i}x_{3i} \\ \sum_{i=1}^{15} x_{1i}x_{2i} & \sum_{i=1}^{15} x_{2i}^2 & \sum_{i=1}^{15} x_{2i}x_{3i} \\ \sum_{i=1}^{15} x_{1i}x_{3i} & \sum_{i=1}^{15} x_{2i}x_{3i} & \sum_{i=1}^{15} x_{3i}^2 \end{bmatrix}$$

$$x^T y = \begin{bmatrix} \sum_{i=1}^{15} x_{1i} y_i \\ \sum_{i=1}^{15} x_{2i} y_i \\ \sum_{i=1}^{15} x_{3i} y_i \end{bmatrix}$$

[illegible]

Now we calculate the mean deviation about mean,

$$\sum_{i=1}^{15} x_{1i}^2 = \sum_{i=1}^{15} X_{1i}^2 - n\overline{X_1}^2 = ???$$

$$\sum_{i=1}^{15} x_{2i}^2 = \sum_{i=1}^{15} X_{2i}^2 - n\overline{X_2}^2 = ???$$

$$\sum_{i=1}^{15} x_{3i}^2 = \sum_{i=1}^{15} X_{3i}^2 - n\overline{X_3}^2 = ???$$

$$\sum_{i=1}^{15} x_{1i}x_{2i} = \sum_{i=1}^{15} X_{1i}X_{2i} - n\overline{X_1}\overline{X_2} = ???$$

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$$\sum_{i=1}^{15} x_{2i}x_{3i} = \sum_{i=1}^{15} X_{2i}X_{3i} - n\overline{X_2}\overline{X_3} = ???$$

$$\sum_{i=1}^{15} x_{1i}y_i = \sum_{i=1}^{15} X_{1i}Y_i - n\overline{X_1}\overline{Y} = ???$$

$$\sum_{i=1}^{15} x_{2i}y_i = \sum_{i=1}^{15} X_{2i}Y_i - n\overline{X_2}\overline{Y} = ???$$

$$\sum_{i=1}^{15} x_{3i}y_i = \sum_{i=1}^{15} X_{3i}Y_i - n\overline{X_3}\overline{Y} = ???$$

$$(x^T x)^{-1} = ??? \text{ [=MINVERSE(K149:M151)]}$$

$$\hat{\beta} = \begin{bmatrix} \widehat{\beta_1} \\ \widehat{\beta_2} \\ \widehat{\beta_3} \end{bmatrix} = ???$$

$$\widehat{\beta_0} = \overline{Y} - \widehat{\beta_1}\overline{X_1} - \widehat{\beta_2}\overline{X_2} - \widehat{\beta_3}\overline{X_3}$$

$$\widehat{\beta}_0 = \log_e A \Rightarrow \hat{A} = e^{\widehat{\beta}_0} = \text{????}$$

$$\hat{\alpha} = \widehat{\beta}_1 = \text{????}$$

$$1 - \hat{\alpha} = \widehat{\beta}_2 = \text{????}$$

$$\widehat{\beta}_3 = -\frac{\hat{\rho}}{2}\hat{\alpha}(1 - \hat{\alpha}) \Rightarrow \hat{\rho} = -\frac{2\widehat{\beta}_3}{\hat{\alpha}(1-\hat{\alpha})} \text{????}$$

The fitted line is

$$Q = \hat{A}[\hat{\alpha}L^{-\hat{\rho}} + (1 - \hat{\alpha})K^{-\hat{\rho}}]^{-\frac{1}{\hat{\rho}}} = \text{????}$$

$$\text{Now, } SS_Y = \sum_{i=1}^{15} Y_i^2 - n\bar{Y}^2 = \text{????}$$

$$SS_R = \widehat{\beta}^T x^T y = \text{????}$$

$$SS_E = SS_Y - SS_R$$

$$R^2 = SS_R / SS_Y$$

Comment: Since the value of $R^2 = \text{????}$, then we can say that the fit is

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$$V(\widehat{\beta}_3) = \hat{\sigma}^2 a_{33} = ???$$

$$\text{and } SE(\widehat{\beta}_1) = \sqrt{V(\widehat{\beta}_1)} = ??? \text{ and } SE(\widehat{\beta}_2) = \sqrt{V(\widehat{\beta}_2)} = ??? \text{ } E(\widehat{\beta}_3) = \sqrt{V(\widehat{\beta}_3)} = ???$$

(c) We want to test the following hypothesis

$$H_0: \beta_1 = 0$$

$$\text{Vs } H_1: \beta_1 \neq 0$$

$$\text{The test statistic is } t = \frac{\widehat{\beta}_1 - \beta_1}{SE(\widehat{\beta}_1)} \sim t_{n-k}$$

Comment:

Again, to test the following hypothesis

$$H_0: \beta_2 = 0$$

$$\text{Vs } H_1: \beta_2 \neq 0$$

$$\text{The test statistic is } t = \frac{\widehat{\beta}_2 - \beta_2}{SE(\widehat{\beta}_2)} \sim t_{n-k}$$

Comment:

Again, to test the following hypothesis

$$H_0: \beta_3 = 0$$

$$\text{Vs } H_1: \beta_3 \neq 0$$

$$\text{The test statistic is } t = \frac{\widehat{\beta}_3 - \beta_3}{SE(\widehat{\beta}_3)} \sim t_{n-k}$$

Comment:

